

Homework (Habanero)

- Q1.** What is the GCF of 12 and 18 using factors?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 9
- Q2.** A pharmacist needs to package 24 pills into bottles of equal size. What is the LCM of 3, 4, and 6 to determine the largest possible bottle size?
(A) 6
(B) 12
(C) 18
(D) 24
(E) 36
- Q3.** What is the GCF of 48 and 60 using prime factorization?
(A) 2
(B) 3
(C) 4
(D) 6
(E) 12
- Q4.** A doctor prescribes a patient to take 15 pills in the morning and 20 pills in the evening. What is the LCM of 15 and 20 to determine the smallest number of days the patient needs to take the pills?
(A) 30
(B) 45
(C) 60
(D) 75
(E) 90
- Q5.** What is the GCF of 27 and 45 using factors?
(A) 1
(B) 3
(C) 5
(D) 9
(E) 15
- Q6.** A biologist is studying the growth of cells. The cells grow in cycles of 8 and 12 hours. What is the LCM of 8 and 12 to determine the time it takes for the cells to complete one full cycle?
(A) 16
(B) 20
(C) 24
(D) 32
(E) 40
- Q7.** What is the GCF of 50 and 70 using prime factorization?
(A) 2
(B) 5
(C) 7
(D) 10
(E) 35
- Q8.** A hospital has 18 nurses and 24 doctors. What is the LCM of 18 and 24 to determine the smallest number of teams that can be formed with an equal number of nurses and doctors?
(A) 36
(B) 48
(C) 72
(D) 96
(E) 144

Open Ended Questions (FRQ)

- Q9.** A pharmaceutical company produces 120 pills per hour. The pills are packaged in bottles of 8 pills each. What is the GCF of 120 and 8 and how many bottles can be filled per hour?
- Q10.** A researcher is studying the effect of a new medicine on a population of 48 patients. The patients are divided into groups of 4, 6, and 8 patients. What is the LCM of 4, 6, and 8 and how many different groups can be formed?

Chef's Tips

- Tip 1: Emphasize the importance of finding the GCF and LCM in real-world problems.
- Tip 2: Encourage students to use prime factorization to find the GCF and LCM.
- Tip 3: Provide examples of how the GCF and LCM are used in the healthcare and biology fields.